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TREND-BASED MOCK PAPER

GATE Petroleum Engineering

Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Petroleum Engineering
Paper:	Mock 04
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Section B: Petroleum Engineering

Q2. [1-MCQ] [1 mark]

First order ODE $dy/dx = y$ solution:

- (A) Ce^x
- (B) Ce^{-x}
- (C) x^2
- (D) $\ln(x)$

Q3. [1-NAT] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

(Numerical Answer Type — Enter the numerical value)

Q4. [1-MCQ] [1 mark]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q5. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q6. [1-MCQ] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q7. [1-NAT] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q8. [1-NAT] [1 mark]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

(Numerical Answer Type — Enter the numerical value)

Q9. [1-MCQ] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q10. [1-MCQ] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q11. [1-MCQ] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q12. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
 - (B) 2
 - (C) 3
 - (D) 0
-

Q13. [2-MCQ] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q14. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q15. [2-MCQ] [2 marks]

$P(|Z| < 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q16. [2-MCQ] [2 marks]

Laplace transform of $e^{-3t} u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q17. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q18. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q19. [2-MCQ] [2 marks]

$d/dx [\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q20. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q21. [2-MSQ] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q22. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q23. [2-MSQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q24. [2-NAT] [2 marks]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q25. [2-NAT] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q26. [2-NAT] [2 marks]

Cauchy-Schwarz inequality states $|\langle u,v \rangle| \leq$:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q28. [2-NAT] [2 marks]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q29. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q30. [2-NAT] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

(Numerical Answer Type — Enter the numerical value)

Q31. [1-NAT] [1 mark]

For Otto cycle, efficiency depends on:

(Numerical Answer Type — Enter the numerical value)

Q32. [1-MCQ] [1 mark]

Carnot engine between 600K and 300K. $Q_{in}=1000J$. Work output:

- (A) 500J
- (B) 333J
- (C) 667J
- (D) 1000J

Q33. [1-NAT] [1 mark]

Max efficiency of Carnot between 500K and 300K:

(Numerical Answer Type — Enter the numerical value)

Q34. [2-MCQ] [2 marks]

$C_p - C_v$ for ideal gas =

- (A) 0
- (B) R
- (C) $2R$
- (D) $R/2$

Q35. [2-MCQ] [2 marks]

Entropy change of universe for reversible process:

- (A) >0
- (B) $=0$
- (C) <0
- (D) Depends

Q36. [2-MCQ] [2 marks]

Isothermal expansion: $V_2=2V_1$, $T=300K$. Work per mole:

- (A) 1729 J
- (B) 0 J
- (C) -1729 J
- (D) -864 J

Q37. [2-NAT] [2 marks]

Air at 27°C, 1 atm compressed to 10 atm isentropically ($\gamma=1.4$):

(Numerical Answer Type — Enter the numerical value)

Q38. [2-MCQ] [2 marks]

Carnot engine between 600K and 300K. $Q_{in}=1000J$. Work output:

- (A) 500J
- (B) 333J
- (C) 667J
- (D) 1000J

Q39. [2-MCQ] [2 marks]

Air at 27°C, 1 atm compressed to 10 atm isentropically ($\gamma=1.4$):

- (A) 534 K
- (B) 507 K
- (C) 300 K
- (D) 600 K

Q40. [1-MCQ] [1 mark]

Mud weight is controlled to:

- (A) Cool bit only
- (B) Prevent kicks and circulation loss
- (C) Speed up drilling
- (D) Lubricate

Q41. [1-MCQ] [1 mark]

Drill bit rotation is provided by:

- (A) Top drive
- (B) Rotary table
- (C) Both
- (D) Downhole motor only

Q42. [2-MCQ] [2 marks]

Drill bit rotation is provided by:

- (A) Top drive
- (B) Rotary table
- (C) Both
- (D) Downhole motor only

Q43. [2-MCQ] [2 marks]

Drill bit rotation is provided by:

- (A) Top drive
- (B) Rotary table
- (C) Both
- (D) Downhole motor only

Q44. [2-MCQ] [2 marks]

Drill bit rotation is provided by:

- (A) Top drive
- (B) Rotary table
- (C) Both
- (D) Downhole motor only

Q45. [1-NAT] [1 mark]

Choke in production system controls:

(Numerical Answer Type — Enter the numerical value)

Q46. [1-MCQ] [1 mark]

ESP (Electric Submersible Pump) is used for:

- (A) Gas lift
 - (B) Artificial lift of liquids
 - (C) Water injection
 - (D) Sealing
-

Q47. [2-MCQ] [2 marks]

Choke in production system controls:

- (A) Flow rate
 - (B) Pressure only
 - (C) Temperature
 - (D) Composition
-

Q48. [2-MCQ] [2 marks]

Choke in production system controls:

- (A) Flow rate
 - (B) Pressure only
 - (C) Temperature
 - (D) Composition
-

Q49. [2-MCQ] [2 marks]

ESP (Electric Submersible Pump) is used for:

- (A) Gas lift
 - (B) Artificial lift of liquids
 - (C) Water injection
 - (D) Sealing
-

Q50. [1-MCQ] [1 mark]

Source rock rich in:

- (A) Carbonate
 - (B) Organic matter
 - (C) Sand
 - (D) Shale
-

Q51. [1-MCQ] [1 mark]

Anticline is a:

- (A) Fault
 - (B) Fold structure
 - (C) Unconformity
 - (D) Reef
-

Q52. [1-MCQ] [1 mark]

Seismic survey uses:

- (A) Sound waves
- (B) Electromagnetic
- (C) Gravity

(D) Magnetic

Q53. [2-NAT] [2 marks]

Anticline is a:

(Numerical Answer Type — Enter the numerical value)

Q54. [2-MCQ] [2 marks]

Anticline is a:

- (A) Fault
- (B) Fold structure
- (C) Unconformity
- (D) Reef

Q55. [1-MCQ] [1 mark]

Darcy's law: $q = ?$

- (A) $kA \frac{\Delta h}{L}$
- (B) $\frac{\Delta h}{L}$
- (C) $kL \Delta h$
- (D) $\frac{\Delta h}{L}$

Q56. [1-NAT] [1 mark]

Darcy's law: $q =$

(Numerical Answer Type — Enter the numerical value)

Q57. [1-NAT] [1 mark]

Material balance equation relates:

(Numerical Answer Type — Enter the numerical value)

Q58. [2-NAT] [2 marks]

Material balance equation relates:

(Numerical Answer Type — Enter the numerical value)

Q59. [2-NAT] [2 marks]

Material balance equation relates:

(Numerical Answer Type — Enter the numerical value)

Q60. [1-MCQ] [1 mark]

Roller cone bit has:

- (A) 1 cone
- (B) 2
- (C) 3
- (D) 4

Q61. [1-NAT] [1 mark]

ESP stands for:

(Numerical Answer Type — Enter the numerical value)

Q62. [1-NAT] [1 mark]

Gamma ray log measures:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-MCQ] [1 mark]

Pressure transient analysis uses:

- (A) Horner plot
- (B) Material balance
- (C) MBAL
- (D) PVT

Q64. [1-NAT] [1 mark]

API gravity ≥ 31.1 is:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-MCQ] [1 mark]

Jack-up rig works in:

- (A) Deep water
 - (B) Shallow water
 - (C) Both
 - (D) None
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	A	1
3	1MCQ	B	1
4	1MCQ	A	1
5	1MCQ	C	1
6	1NAT	A	1
7	1NAT	C	1
8	1MCQ	A	1
9	1NAT	B	1
10	1MCQ	B	1
11	1NAT	A	1
12	1MCQ	B	1
13	1MCQ	A	1

14	1MCQ	C	1
15	1MCQ	B	1
16	1MCQ	A	1
17	1NAT	A	1
18	1MCQ	B	1
19	1NAT	A	1
20	1NAT	B	1
21	1MCQ	A	1
22	1NAT	A	1
23	1MCQ	C	1
24	1NAT	A	1
25	1MCQ	A	1
26	1NAT	B	1
27	1MCQ	B	1
28	1MCQ	B	1

29	1MCQ	B	1
30	1NAT	A	1
31	2MCQ	B	2
32	2MCQ	B	2
33	2MCQ	B	2
34	2MCQ	A	2
35	2MCQ	C	2
36	2MCQ	A	2
37	2MCQ	A	2
38	2MCQ	A	2
39	2MCQ	A	2
40	2MCQ	C	2
41	2MCQ	A	2
42	2MCQ	A	2
43	2MSQ	B	2

44	2MCQ	B	2
45	2NAT	B	2
46	2MSQ	A	2
47	2NAT	C	2
48	2NAT	C	2
49	2NAT	A	2
50	2MCQ	A	2
51	2NAT	B	2
52	2MCQ	C	2
53	2NAT	A	2
54	2MCQ	C	2
55	2MCQ	A	2
56	2MCQ	C	2
57	2MCQ	A	2
58	2MCQ	B	2

59	2NAT	B	2
60	2NAT	B	2
61	2MCQ	C	2
62	2MCQ	A	2
63	2NAT	B	2
64	2MCQ	B	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Differential Equations | Difficulty: difficult

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Petroleum Exploration | Topic: Geology | Difficulty: moderate

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Reservoir Engineering | Topic: Fluid Flow | Difficulty: easy

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Drilling Engineering | Topic: Bits | Difficulty: easy

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Production Engineering | Topic: Artificial Lift | Difficulty: easy

Q7.

Answer: C

Solution:

Correct Answer: C

Subject: Petroleum Formation Evaluation | Topic: Logging | Difficulty: difficult

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Well Testing | Topic: Analysis | Difficulty: moderate

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Petroleum Chemistry | Topic: Crude | Difficulty: difficult

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Offshore Drilling | Topic: Rigs | Difficulty: difficult

Q11.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q12.

Answer: B

Solution:

Anticline: upward convex fold. Good petroleum trap.

Subject: Petroleum Exploration | Topic: Petroleum Exploration | Difficulty: difficult

Q13.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q14.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q15.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = Cx$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = Cx$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = Cx$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q16.

Answer: A

Solution:

Seismic: generates and records sound waves reflected by subsurface layers.

Subject: Petroleum Exploration | Topic: Petroleum Exploration | Difficulty: difficult

Q17.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=xv$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q18.

Answer: B

Solution:

Mud weight: balance formation pressure (prevent kicks) and prevent fracture (lost circulation).

Subject: Drilling | Topic: Drilling | Difficulty: moderate

Q19.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q20.

Answer: B

Solution:

Otto: $\eta = 1 - r^{-(1-\gamma)}$. Depends on compression ratio r and γ .

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q21.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q22.

Answer: A

Solution:

$q = -k \cdot A / \mu \cdot dp/dL$. Steady linear flow: $q = k \cdot A \cdot dp / (\mu \cdot L)$.

Subject: Reservoir Engineering | Topic: Reservoir Engineering | Difficulty: moderate

Q23.

Answer: C

Solution:

Rotation: surface (top drive or rotary table) and/or downhole motor (mud motor).

Subject: Drilling | Topic: Drilling | Difficulty: easy

Q24.

Answer: A

Solution:

Choke: restricts flow to control production rate and upstream pressure.

Subject: Production | Topic: Production | Difficulty: easy

Q25.

Answer: A

Solution:

$\eta = 1 - 300/600 = 0.5$. $W = 0.5 \cdot 1000 = 500J$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q26.

Answer: B

Solution:

Material balance: OOIP, cumulative production, reservoir pressure evolution.

Subject: Reservoir Engineering | Topic: Reservoir Engineering | Difficulty: moderate

Q27.

Answer: B

Solution:

ESP: multistage centrifugal pump at bottom of well. Artificial lift for liquid production.

Subject: Production | Topic: Production | Difficulty: easy

Q28.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q29.

Answer: B

Solution:

$P(A|B) = P(A \cap B) / P(B) = 0.2 / 0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q30.

Answer: A

Solution:

$\eta = 1 - 300/500 = 1 - 0.6 = 0.4 = 40\%$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q31.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q32.

Answer: B

Solution:

Mayer's relation: $C_p - C_v = R$ for ideal gas.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q33.

Answer: B

Solution:

Reversible: $\Delta S_{\text{univ}} = 0$. Irreversible: $\Delta S_{\text{univ}} > 0$ (second law).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q34.

Answer: A

Solution:

$L\{t^*f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q35.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q36.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q39.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4}$ approx 0.908.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q41.

Answer: A

Solution:

$W = nRT \ln(V_2/V_1) = 8.314 * 300 * \ln(2) = 2494.2 * 0.693 = 1729 \text{ J}$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q42.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q43.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: B

Solution:

$T_2/T_1 = (P_2/P_1)^{(\gamma-1)/\gamma} = 10^{(0.286)} = 1.93$. $T_2 = 300 \times 1.93 = 579\text{K}$. Closest: 507K (using exact calculation).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q46.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q49.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q50.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q51.

Answer: B

Solution:

Anticline: upward convex fold. Good petroleum trap.

Subject: Petroleum Exploration | Topic: Petroleum Exploration | Difficulty: moderate

Q52.

Answer: C

Solution:

Rotation: surface (top drive or rotary table) and/or downhole motor (mud motor).

Subject: Drilling | Topic: Drilling | Difficulty: moderate

Q53.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q54.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q55.

Answer: A

Solution:

Choke: restricts flow to control production rate and upstream pressure.

Subject: Production | Topic: Production | Difficulty: easy

Q56.

Answer: C

Solution:

Rotation: surface (top drive or rotary table) and/or downhole motor (mud motor).

Subject: Drilling | Topic: Drilling | Difficulty: moderate

Q57.

Answer: A

Solution:

Choke: restricts flow to control production rate and upstream pressure.

Subject: Production | Topic: Production | Difficulty: moderate

Q58.

Answer: B

Solution:

Anticline: upward convex fold. Good petroleum trap.

Subject: Petroleum Exploration | Topic: Petroleum Exploration | Difficulty: easy

Q59.

Answer: B

Solution:

Material balance: OOIP, cumulative production, reservoir pressure evolution.

Subject: Reservoir Engineering | Topic: Reservoir Engineering | Difficulty: moderate

Q60.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q61.

Answer: C

Solution:

Rotation: surface (top drive or rotary table) and/or downhole motor (mud motor).

Subject: Drilling | Topic: Drilling | Difficulty: moderate

Q62.

Answer: A

Solution:

$\eta = 1 - 300/600 = 0.5$. $W = 0.5 \cdot 1000 = 500\text{J}$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q63.

Answer: B

Solution:

Material balance: OOIP, cumulative production, reservoir pressure evolution.

Subject: Reservoir Engineering | Topic: Reservoir Engineering | Difficulty: difficult

Q64.

Answer: B

Solution:

$T_2/T_1 = (P_2/P_1)^{(\gamma-1)/\gamma} = 10^{(0.286)} = 1.93$. $T_2 = 300 \cdot 1.93 = 579\text{K}$. Closest: 507K (using exact calculation).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q65.

Answer: B

Solution:

ESP: multistage centrifugal pump at bottom of well. Artificial lift for liquid production.

Subject: Production | Topic: Production | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 29 questions, 48 marks
Thermodynamics: 9 questions, 15 marks
Drilling: 5 questions, 8 marks
Production: 5 questions, 8 marks
Petroleum Exploration: 5 questions, 7 marks
Reservoir Engineering: 5 questions, 7 marks
General Aptitude: 1 questions, 1 marks
Drilling Engineering: 1 questions, 1 marks
Production Engineering: 1 questions, 1 marks
Petroleum Formation Evaluation: 1 questions, 1 marks
Well Testing: 1 questions, 1 marks
Petroleum Chemistry: 1 questions, 1 marks
Offshore Drilling: 1 questions, 1 marks

Difficulty Distribution:

Easy: 18 | Moderate: 29 | Difficult: 18

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.